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## ABSTRACT:

The **Information Technology & Information Technology Enabled Services (IT-ITeS)** sector is a field which is undergoing rapid evolution and is changing the shape of business standards. This sector includes software development, software testing, consultancies, software management, online services and business process outsourcing (BPO). As we all know technology is revolutionizing the way we do business and making high-tech approaches an integral part of our lives. Back in 1990's when no one talked about the software quality, in a modern era, the industry today has awakened to the fact that testing is more important than programming. Testing apps requires a bigger budget for tools and resources compared to programming. Every organization today is hunting for the best possible talent. Earlier software developers used to write code and testers checked it for quality. This won't suffice today. Another change was increasing demand for more frequent software releases. This was the result of many factors, like the commercialization of software, and businesses' desire to keep customers happy by providing new and updated products on a consistent basis. These changes raised the stakes for software testing. Releasing software that worked on any PC required careful configuration testing of the many possible environment variables. At the same time, the fact that users had come to expect more frequent releases meant that programming teams had to optimize their testing processes, so they could deliver faster. Engineers who are a hybrid of software developer, DevOps adoption, tester combined automated and manual testing and more are in high demand. From small startups to mega-tech companies, great SDETs (Software development engineering in test) are highly sought after.

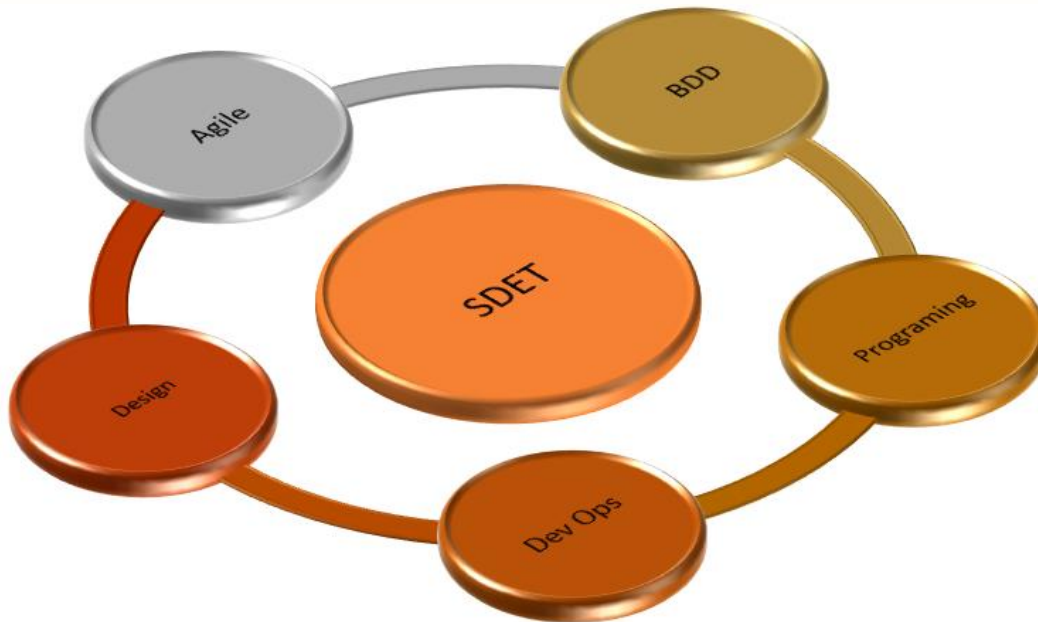
**Keywords** - IT-ITeS, SDETs, BPO, software development, software testing, consultancies, software management.

## AUDIENCE AND PURPOSE:

This white paper discusses the current software testing industry situation and a requirement of SDET's that best fits industry needs and intended to educate the target audience, that have a basic knowledge of software test industry and want to know the future of the software testing industry. Today when the business has begun rising and the organizations have propelled their advancements ahead, quality is the fundamental worry that gives the right incentive with no re-work or higher costs required for the Software Development. In prior days, the software engineer used to write the code and the tester was compelled to test it for quality; which was only the Black-box Testing. But now Software testing is performed by Software Test Engineers (STE) and Software Development Engineers in Test (SDET).



### INTRODUCTION



*“Program testing can be used to show the presence of bugs, but never to show their absence!”*

– Edsger Dijkstra, Dutch computer scientist.

The industry today has awakened to the fact that testing is actually more important than programming. Testing apps requires a bigger budget for tools and resources compared to programming. Every organization today is hunting for the best possible talent. Earlier software developers used to write code and testers checked it for quality. This won't suffice today.

Software Development Engineer in Test (SDETs) are skilled professionals who are adept in the arena of both QA Engineering and Software Development.

Though it's great to have someone with skills and expertise that are high in demand, it also creates a bit of confusion as to what really comprises the duties and responsibilities of SDETs. A regularly faced ambiguity is regarding the difference between an SDET and a QA Engineer.



## AUTOMATION IS KEY TO QA PARADIGM



Automation has changed the face of software development. Today we no longer provision boxes manually. We treat computing infrastructure as code. The same is true for deployment. Gone are the days of weekend war rooms where a bunch of sysadmins got together with cases of Coke and bags of chips to install operating systems and applications. Hotfixes are no longer done by hand copying files via SSH on a case by case basis. We create and destroy environments on a whim. Our releases are run by code that makes deployment continuous and constant.

The same dynamic is happening in QA. When it comes to testing in the modern enterprise, the simple fact is either you're an SDET or you're not. Yes, it's true that more advanced testing such as complex, multi-system performance testing requires some manual intervention. But, for typical UI and functional tests, these are -- and must be -- handled by automation, otherwise, the competition will prevail.

But still, a set of myths linger on in the minds of many non-technical executives. The myths -- though not ubiquitous, but prevalent -- are that anybody can be taught to program and that any entry-level manual

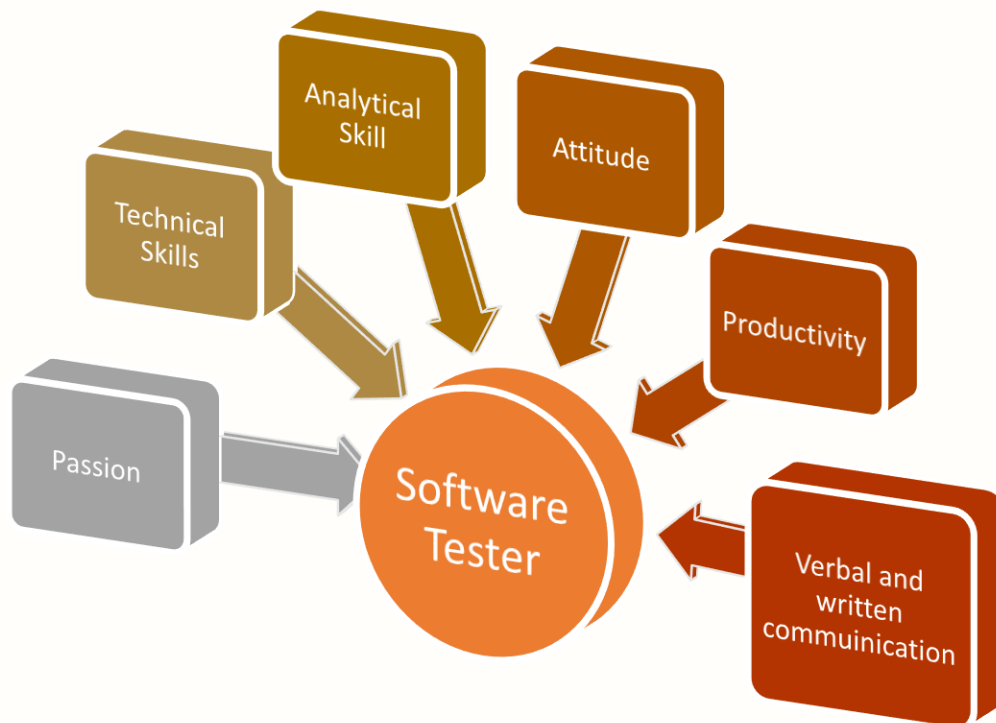
## THE NEED OF SDET'S



Armed with specialized testing knowledge of multiple tools, techniques, best practices, and processes, SDETs today have become a crucial part of development ecosystems. Based on their development experience, knowledge of technical architecture and design, and their programming skills SDETs are required to write a code to test the code written by developers. In addition, they are also required to write unit tests and also perform white box testing.



The position is a consolidated package of the Software Development Life Cycle and in addition Test Life Cycle both. SDET designs, plans, creates, streamlines and also tests the code utilizing automation. An SDET can consider the possibilities the application code can fail and plan his test structure appropriately to guarantee the durability of the product.



Some say the move is simply to draw in the engineering talent. The common misconception is ‘There is no vocation in Software Testing’, bringing about a deficiency of skilled workforce picking the testing field. Everybody needs to be an engineer first.

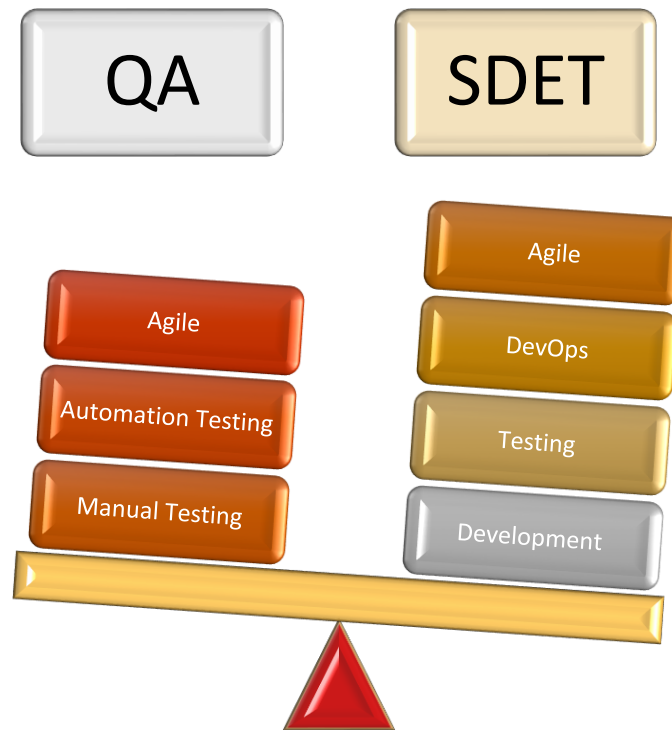
Also, the **Stress on Automation** testing are increasing day by day and is certainly essential, however, executing it without a definitive target into viewpoint, can be grievous. Eventually, there are investments included and groups are responsible for the outcomes.

Automation testing can’t ever trade the requirement for Manual testing, and companies are not willing to employ two distinctive kinds of testers – how about they hire SDETs?

So, SDET doesn’t mean software testing is leaving. It just implies that we are shifting from a pure manual testing to a more technical reality of testing a product. SDET isn’t only a test professional with advanced level programming skills, yet a professional who can create testing frameworks and tools to help the manual testing process.



## SDETS VS TRADITIONAL QA ENGINEERS

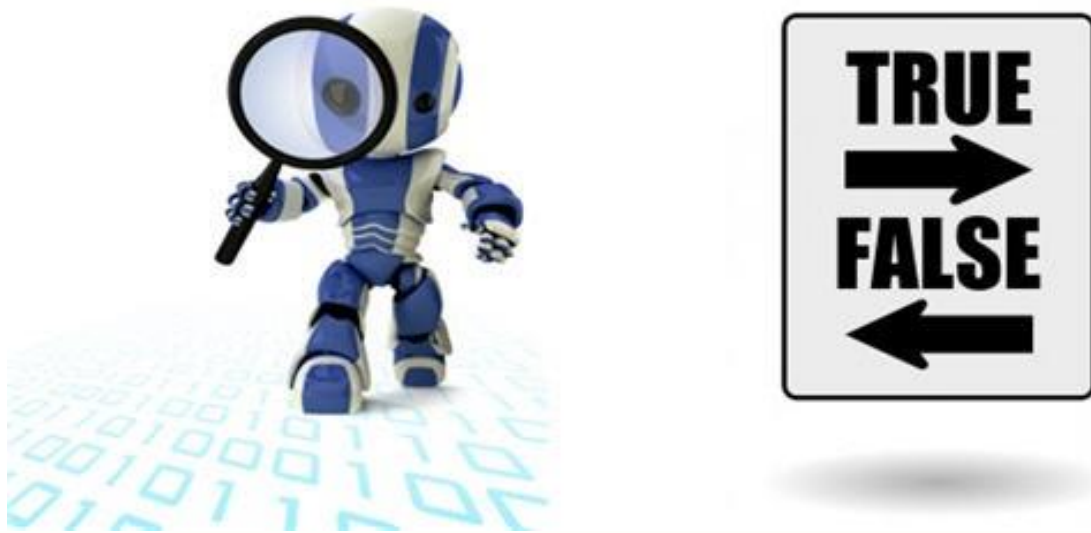


While the roles of SDETs may seem very similar to those of QA Automation Engineers, with even most of the tools and language expertise required being same (Selenium, Java, and Jenkins), there are certain very obvious and clear differences between the two roles.

An SDET, in layman terms, is a developer who instead of working in the product development team, works as part of the test team. SDETs are responsible not only for writing code but are required to test the code as well. SDETs are required to continuously write, test, and fix the code they write. Their roles and responsibilities are based on the Agile lifecycle model. SDETs usually are professionals who have very strong analytical, technical, and problem-solving skills.

On the other hand, QA Engineers are testers who do not need to have any programming experience as they usually are not exposed to the code. This clearly creates a demarcation between the roles and responsibilities of SDETs and QA Engineers.

On a high-level side, SDET can be viewed as a software developer and likewise a tester though he will be a member of the testing stage and not of the development stage of the software development lifecycle.



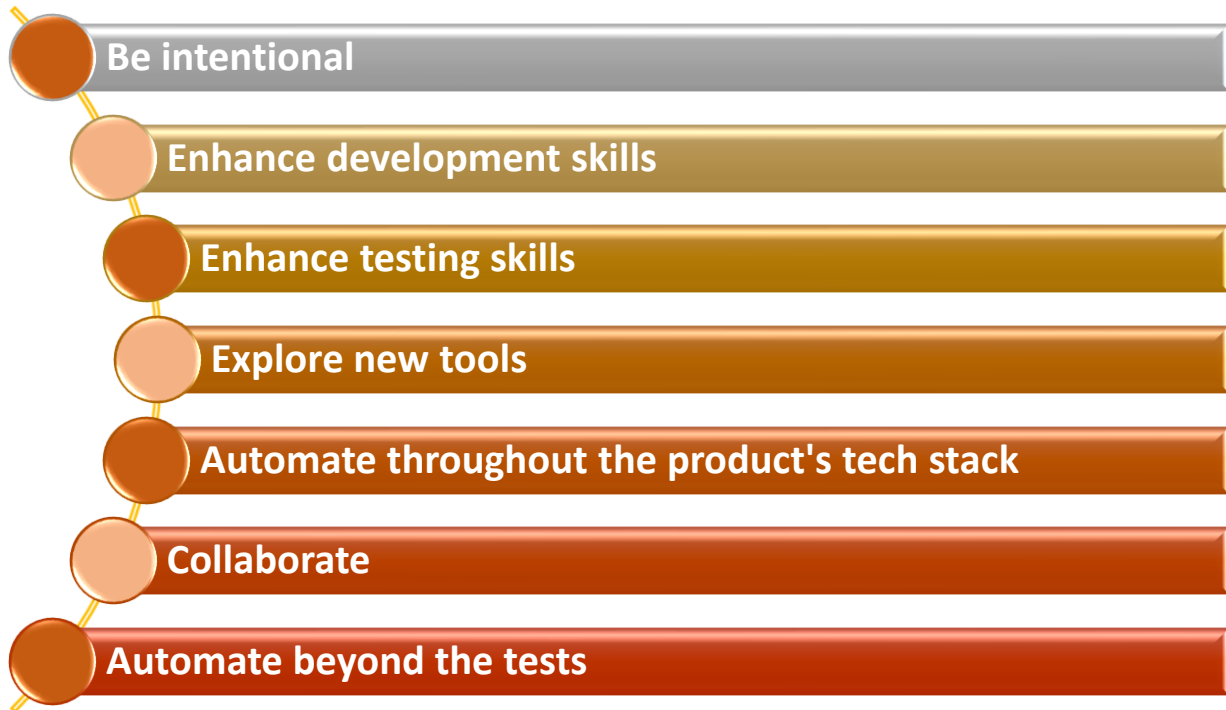
Let's Have a Look at "Rex Black" Former President of International Software Testing Qualification Board (ISTQB) explaining the Difference Between a Tester and a SDET

- SDET is an extremely skilled resource with development and testing skills; on the contrary, the tester is required in planning and performing the test cases either by automation or manually.
- Testers are sources with restricted programming abilities and they are much centered upon functional testing or black box. However, SDETs are skilled resources with high-grade programming skills and perform the job of the tester i.e. doing white box testing and additionally as a developer in test automation.
- SDET is highly skilled in software development; they can partake in test automation tool development and may create it for general usage. Testers are not supposed to build test automation tools, they may utilize these test automation tool to automate the test cases needed for their project.
- Unlike manual testers, SDET is assumed to have field experience in order to participate in creating the test cases. All responsibilities of a tester are required from SDET.
- SDET is supposed for more jobs over general testing role such as performance testing, automated generation of test data, security testing, test environment setup, etc.

Usually, it's recognized in a company, QA or tester who have developed the test automation framework are advanced to a new role as SDET and can engage in a review or design of software application or different test automation system.



## HABITS OF HIGHLY EFFECTIVE SDETS



Software developers in test (SDETs), engineers who are a hybrid of software developer and tester, are in high demand. From small startups to mega-tech companies, great SDETs are highly sought after.

But what exactly makes a great SDET? As someone who regularly interviews people to fill SDET positions, I can attest that finding a top-notch one is not easy. This is due not to a shortage of applicants, but to a shortage of applicants who possess the traits that make for an awesome SDET.

Here are some habits that I see in highly effective SDETs.

### 1. **BE INTENTIONAL**

Inexperienced SDETs are of the mindset that they must automate "all the things," or at least all of the things that can be automated. Their goal is to eliminate as much manual testing as possible.

But SDETs who have participated in successful automation projects know that blindly automating tests can cause more problems than it solves. Effective SDETs are very intentional about what they automate and ensure that their selections align with their end goal.



### 2. ENHANCE DEVELOPMENT SKILLS

Many aspiring SDETs believe that learning how to use popular test automation APIs is enough to be successful. That is certainly a starting point, but to be truly effective, more software development skills are required.

Having a firm grasp of programming logic and concepts certainly makes for a more effective SDET. All of the mature automation projects I've been part of have used object-oriented programming, decision structures, repetition structures, and data structures.

Clean coding practices, as well as the use of design patterns, are critical for maintenance as well as reusability and extendibility.

### 3. ENHANCE TESTING SKILLS

While it's important that SDETs be strong programmers, the most effective SDETs are strong testers as well. If the most beautifully written test automation code is not adequately validating the expected behavior, then it's worthless. This type of script will almost always report success because it's not actually validating anything—or at least not what it should be. This sounds really trivial, but I've caught this negligence in many code reviews.

One major reason for this is that the SDET doesn't understand how meticulous testing truly is and doesn't capture all of the many verification points a tester processes before declaring a test successful.

Another reason for such oversight is that it takes great skill to be able to balance the developer and tester mindsets that this hybrid role requires. When developing the framework for a particular test, it's easy to get caught up in the act of creation, and once the test is completed, there is a sense of accomplishment. Effective SDETs can switch their brain to testing mode within this same session.

### 4. EXPLORE NEW TOOLS

Mastering an automation tool is a great feat; however, effective SDETs know that not all challenges can be solved with the same tool. Because of this, they continuously evaluate and add to their collection of tools.

Not only do they learn how to use several tools, but more important, they also know in which contexts each tool is best utilized. Scroll through any test automation forum on the web, and it won't take long before you find someone asking about how to use a UI automation tool such as Selenium WebDriver to automate web service testing or about which tool to start studying to learn automation.

Effective SDETs are able to analyze the problem that needs to be solved and choose which tools are best for that particular problem, as opposed to trying to solve every problem with the one tool that they know how to use.



### 5. AUTOMATE THROUGHOUT THE PRODUCT'S TECH STACK

Effective SDETs are well aware of the costs and risks of automating at the UI layer of the application. Because of this, they do it sparingly and only when necessary. They understand that to effectively automate tests, it's wise to also understand the tech stack of the system under test so that more reliable layers can be utilized in automation.

They look for seams within the application that will enable faster, targeted testing, and they aren't afraid to code tests at the unit and service layers. Again, having knowledge of multiple tools enables SDETs to automate outside of the box of UI automation.

### 6. COLLABORATE

Great SDETs realize that automation is not their responsibility alone. They are not blindly automating tests, but are instead strategic in their approach. Part of figuring out the strategy is to collaborate with other key players.

This includes those who are doing manual or exploratory testing. These types of testers usually have deep domain knowledge of the application and broad thoughts on how to approach testing it. Their input is extremely valuable in determining which tests should be automated.

Developers are also great resources to collaborate with. Not only can they contribute to the test automation code base, but they can also help SDETs understand what additional layers and seams are available.

Business analysts and product owners also provide value to SDETs by helping them understand why certain features are being built and how they will be used. They can also provide historical data of how customers use existing features. All of this information is valuable in determining an automation strategy.

### 7. AUTOMATE BEYOND THE TESTS

While most SDETs mostly focus on automating tests, the great ones also use their skills to automate other tasks as well. They regularly seek out opportunities to make the software development process more efficient by eliminating the boring, repetitive tasks that distract humans from doing the really important and interesting things. Data generation, environment setup, and log parsing are all time-consumers that can be automated to assist with testing efforts.



## TECHNICAL SKILLS REQUIRED FROM SDET



- Understanding of C#, Java, .NET, or other software languages.
- Knowledge of Object-Oriented Design.
- Experience in managing AGILE + DevOps process administration methodology.
- SDET ought to have the exposure to Behavior Driven Development.
- SDET ought to have knowledge about different test method and similar tools like MS Test, TestNG, JUnit, Selenium WebDriver, and so forth.
- Skilled to discover thresholds and bottlenecks in existing code with the cooperation of automation tools.
- A minimum of three years of systems/server-side programming in a professional setting
- At least 1 year of testing experience in a professional setting and/or an interest in this kind of development.

## RESPONSIBILITIES OF AN SDET

- SDET must be capable to perform Test Automation and setting up systems on multiple application platforms like Web, Desktop, and Mobile.
- SDET reviews customer queries referred to by the technical support members.
- SDET further works with deployments teams and fixing any level of problems for the system.
- SDET requires managing technical communications with associates to understand the systems or APIs of the clients.
- He/ She is proficient to create various test environments and acceptance tests.



- SDET builds and controls bug reports and interact with the team.
- An SDET must be able to note down all the testing actions which he operates over daily or weekly into a Report.
- An SDET must comprehend the flow of the code, on how it interacts with every different component.
- SDET must also be able to control, set up, and perform test automation frameworks.
- An SDET must possess superior debugging abilities to discover out the causes behind the ineffective code, breakdown of test cases, memory leakages etc.
- SDET must analyze and disintegrate heterogeneous software systems and cooperate with and motivate others to enhance the overall design.
- SDET should research developing test tools, methodologies, trends and upgrade existing practices and processes.
- SDET has to take part in product design discussions to offer input on the testability of functional elements, product designs, and planning or estimating.
- SDET is responsible for managing quality on the production release.
- SDET should also interpret customer demands and product objectives for relevant tests.

### ORGANIZATIONS DEMAND FOR SDET

There is a trend amidst organizations to embrace new methodologies and technology. Hence, a Software Development Engineer in the Test is very much on demand. SDET professionals are largely required in the following sectors –





## STORIES OF IT LEADERS AND COMPANIES:

At least as of April 2011, Facebook had no employees who were dedicated to QA or otherwise performed QA as their primary job responsibility. There were some employees who do some vaguely QA-like things, but this was a small part of their job.

Wait, **nobody is “dedicated to QA”** at Facebook? Maybe this isn’t the similarity I was thinking of. Surely, Google has software testers. Here we go, it looks like James Whittaker will clear things up for me over at the Google Testing Blog: [ <https://testing.googleblog.com/2011/01/how-google-tests-software.html>]

Product teams can’t place too big a bet on (testers) and must keep their quality house in order. Yes, that’s right: at Google it’s the product teams that own quality, not testers. Every developer is expected to do their own testing. The job of the tester is to make sure they have the automation infrastructure and enabling processes that support this self-reliance. Testers enable developers to test.

What does Facebook have to gain from having “no employees who are dedicated to QA,” and, why wouldn’t testers actually do testing at Google? Says Priestly: By paying less attention to quality, Facebook has been able to focus on other things, like making the company a fun place to work at and how to that can attract and retain talented engineers.

After giving myself a minute to lay down and try to comprehend what I’d just read, I recalled that Tricentis founder, Wolfgang Platz, recently wrote about this same topic—but from a differing viewpoint—for InfoWorld. In “Agile developers test, but aren’t testers,” [ <https://www.infoworld.com/article/3259386/agile-developers-test-but-aren-t-testers.html> ] Platz points out that “agile adoption has blurred the historical distinction between testers and developers, and that’s a good thing.” Platz defends the murkiness of these roles by saying that successful agile development will result in developers testing more often than in the past. However, “If your testing is primarily comprised of bottom-up tests designed by engineers, you’re likely to overlook critical issues,” and you’ll ultimately “threaten to undermine the goals of agile.”

The wide variety of views on how much testing should developers perform, and, likewise, how much programming knowledge should be required of testers, has resulted in an oftentimes controversial debate in the tech industry. A simple Google search of “Should developers test?” and “Should testers learn to code” produces over 10 million search results going back nearly ten years.

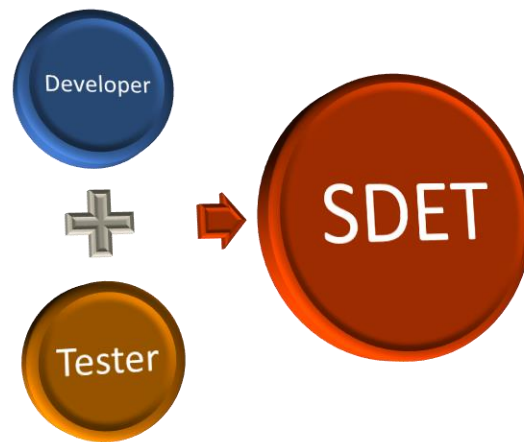
In an effort to give all parties the opportunity to share their thoughts on this matter, Tricentis is hosting what we’re calling “Development.” [ <http://innovate.tricentis.com/Webinar-The-Great-Debate.html> ] Accenture Managing Director: Jeff Wilkinson, Electric Cloud CTO: Anders Wallgren, and Tricentis founder: Wolfgang Platz will provide their own unique takes on who should ultimately be responsible for software quality, and how agile and DevOps have shifted expectations and skill sets.



## CONCLUDING THOUGHTS

Furnished with the specific testing knowledge of various tools, strategies, best practices, and procedures SDET in test today has turned into a critical part of a development environment. As per their knowledge of technical architecture and design, development experience, and their programming aptitudes software developer in test is needed to write a code to test the code composed by the programmers. SDET is not just more than a tester. software development engineer in test is a combination of the tester as well as developer.

The result – let's call it **SDET = developer + tester**, right?



Therefore, the role of the software development engineer in test is very impressive as well as challenging as it strives to deliver the software with the greatest possible quality. To conclude, it's felt that the software development engineers in the test might place himself/herself as an all-rounder. SDET are not just more than a tester. SDET is a mix of developer as well as tester who has exposure to project management, interacts and understands end user software requirements, knows product or domain knowledge, participates in product or software designing, knows how to code and build test automation tools. This all in one kind of skill set make the SDET role very unique and high demand in present software industry.



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